

CHUN-MIN YU

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EDUCATION

National Tsing Hua University (NTHU)

B.S. in Computer Science

Sep 2022 - June 2026

Hsinchu, Taiwan

· GPA: 3.86/4.3

· TOEIC: 880; TOEFL iBT: 100

· **Relevant Coursework:** Machine Learning, Deep Learning, NLP, Compiler Design, Computer Vision, Operating Systems, Embedded Systems, Algorithms, Artificial Intelligence

National Tsing Hua University (NTHU)

M.S. in Computer Science

Currently Enrolled

Hsinchu, Taiwan

SKILLS & EXPERTISE

EXPERTISE: NLP, LLM, Computer Vision, Multimodal learning

PROGRAMMING LANGUAGES: C, C++, C#, Python, Verilog, Dart

FRAMEWORKS & TOOLS: PyTorch, Linux, HuggingFace Transformers, Flutter, Firebase

RESEARCH & EXPERIENCES

NLPSA Lab

Research Assistant, Advisor: Prof. Lun-Wei Ku

June 2025 – Present

Multimodal learning in NLP & Computer Vision(WACV 2026 Demo as First Author)

- Developed an end-to-end system for **athletic potential assessment** via alignment-based analysis of fundamental human movements.
- Implemented a **two-phase training strategy**: utilized HumanML3D for cross-modal feature alignment of motion tokens, followed by biomechanical description tuning using **LLaMA-3 with LoRA**.

Human Motion Analysis & Prediction

- Designed a jump height prediction pipeline using **attention over joint tokens** to model spatio-temporal dynamics, achieving **90.1%** frame-level accuracy.
- Collaborated on research to build a **reinforcement learning framework (GRPO)** on top of a text-conditioned **diffusion** model for **motion generation**.

IKM Lab

Research Assistant, Advisor: Prof. Hung-Yu Kao

April 2024 – Dec 2025

Attention-related Mechanism Research

- Proposed an **attention-skip mechanism** for transformers to improve performance. Awarded **2nd Place** in the Computer Science, and **3rd Place** in the EECS Undergraduate Project Competitions at NTHU.
- Achieved **consistent performance improvements(0.2-1.1%)** on the GLUE, IMDB and SQuAD benchmark.

DataLab

Research Assistant, Advisor: Prof. Shan-Hung Wu

Jan 2026 – Present

LLM Continual Learning Research

- Investigating methods to **transition LLM in-context learning (ICL) into persistent memory**, addressing context window constraints and mitigating catastrophic forgetting.
- Designed a novel **context-compressing attention architecture** to mitigate context window constraints and catastrophic forgetting, with empirical evaluations underway.

PROJECTS

Action Recognition with Embedded Machine Learning

Jan 2025 – Feb 2025

- Developed an **embedded ML system** on Nuvoton M032KG to classify actions via accelerometer data.
- Deployed a real-time CNN structure within limited memory, achieving **81.4%** accuracy.

Generative Semantic Augmentation for Robot Vision

Sep 2025 – Jan 2026

- Engineered a two-stage generative augmentation pipeline for temporally consistent object swapping, integrating **SAM-I2V** for precise segmentation and **Wan 2.1 (DiT)** for latent inpainting.
- Developed a VideoQA module using **YOLOv8n** and **Qwen2.5-VL** to generate structured interaction descriptions for semantic validation of the synthesized data.